Small Models, Careful Edits: On-Device Language Models for Scientific Writing

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Abstract

Language models are now routinely used to revise scientific prose, yet most tools send unpublished manuscripts to remote servers. We study whether a small model running entirely on a laptop can make useful, verifiable edits to LaTeX source. Across 80 edits, the on-device model preserved every citation, reference and label in 98% of cases.

1 Introduction

Writing assistants based on large language models have become common in research workflows [Smith and Lee, 2024]. Their usefulness, however, comes at a cost: the text of an unpublished paper leaves the author's machine. We ask a narrower question than prior work. Can a model small enough to run on a laptop make edits that an author would accept, without breaking the LaTeX that surrounds the prose?

2 Method

Given a paragraph p and an instruction a, the model proposes an edit $\hat{p}$. We accept $\hat{p}$ only if it keeps the set of keys $K(p)$ and every math segment of p intact:

$$\mathrm{accept}(p, \hat{p}) = \mathbb{1}\left[\, K(\hat{p}) = K(p) \;\wedge\; M(p) \subseteq M(\hat{p}) \,\right]. \tag{1}$$

Edits that fail Eq. (1) are retried once with the failure named in the prompt.

3 Results

Table 1 summarises the four editing actions. Rewriting and shortening never dropped a key; the two failures changed units inside an inline formula.

Action	Intact	Median time (s)
Rewrite	20/20	3.2
Shorten	20/20	2.9
Fix grammar	19/20	3.6
Formalize	19/20	3.8

Table 1: Edits that kept all LaTeX intact, per action.

References

Jane Smith and Min Lee. Writing assistants in the research workflow. *Journal of Scholarly Tools*, 2024.